

- 6-DOF custom robotic arm designed in Fusion 360 and exported to URDF.
- Operating Environment: ROS 2 Jazzy, Gazebo Harmonic, Linux.
- Current Issue: Gripper mimic joints act individually in MoveIt 2 Setup Assistant and conflict with Gazebo Harmonic physics engine.
- Planned Resolution: Define mimic fingers as "Passive Joints" in MoveIt 2 and only include the active driver joint in the planning group.
- Planned Resolution: Adjust `ros2_control` hardware interfaces to only send commands to the active joint and update Gazebo configurations to handle mimic visual states/physics.
- Completed Task: Set up a dedicated `robot_arm_gazebo` package with Launch files (Gazebo + RViz).
- Resolved Issue: Fixed classic Gazebo plugins `libgazebo_ros_control.so` to modern `gz_ros2_control-system` for Harmonic.
- Resolved Issue: Exported `GZ_SIM_RESOURCE_PATH` to resolve `model://` URI mesh not found errors.
- Resolved Issue: Changed Gazebo Harmonic physics engine to `gz-physics-bullet-featherstone-plugin` to correctly support URDF `<mimic>` joints (Dart engine lacks mimic support).
- Resolved Issue: Fixed mesh loading in Gazebo by converting `package://` URIs to `model://` in `robotic_arm.xacro` and correctly exporting the workspace install path.
- Completed Task: Modernized `<ros2_control>` tags in `robotic_arm.trans`, successfully isolating passive mimic joints from active driver joints.
- Discovered Issue: Gazebo throws warnings for legacy Ogre `Gazebo/Silver` materials. Not a fatal error (Harmonic uses defaults), but good to note.
- Next Task: Create `controllers.yaml` and integrate controller execution to the launch file before running MoveIt 2 Setup Assistant.